
Transformer based Time Series Latent Diffusion Model for Bio-signal generation

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Abstract

The generation of high-fidelity synthetic Electrocardiogram (ECG) signals has become increasingly important for expanding training datasets in medical research, enabling improved machine learning models while safeguarding patient privacy. Generative Adversarial Networks (GANs) are a traditional architecture used to achieve good results in this domain but suffer from issues like mode collapse and poor performance with limited training data. Diffusion models are the recent breakthrough in generative AI, providing more stability and higher diversity in generated samples when compared to GANs. Thus, we propose a Transformer-based time series diffusion architecture that leverages the strengths of diffusion processes and self-attention mechanisms to produce realistic and varied ECG waveforms. We show that even with limited number of samples, Transformer based Diffusion models are better at producing viable synthetic ECG signals. **Link to our github repository.**

1 Introduction

There is a significant surge in the usage of Artificial Intelligence tools in the medical field. These tools help professionals in various subfields diagnose patients better and more confidently. Mayo Clinic has developed a new screening tool for people with a condition called left ventricular dysfunction that does not have obvious symptoms. The tool was able to correctly identify the condition 93% of the time compared to a mammogram with only 85% precision. Such AI tools developed at the Mayo Clinic [1] are also being used in Apple Watches to detect low ventricular ejection function. Therefore, access to a comprehensive ECG database with varied real-world signals captured during clinical practice is necessary. The standard data sets for ECG signals are highly biased as a very low percentage of people are diagnosed with conditions such as the one mentioned above. It is a standard belief that models are as good as the data used to train it, thus there is a need to balance data with synthetic generation.

There are also other uses for ECG data and synthetic data generation. ECG signals are excellent bio-markers and can be used for biometric authentication. [2] shows such a system of biometric authentication. Using generated data can allow to make these systems robust.

Since we are dealing with such sensitive data, there is a need to have good and stable synthetic signals. In recent years, diffusion models have emerged as a powerful generative framework, demonstrating superior performance in generating high-fidelity samples across domains such as images, audio, and time-series data. When combined with Transformer-based architectures, which are highly adept at capturing long-range dependencies, this approach becomes especially well-suited for modeling 1D signals like ECG, which inherently exhibit both global rhythm and localized features.

In this work, we propose a framework that leverages LDMs along with Transformer-based architectures to generate synthetic ECG signals. We build upon a baseline model, Diffusion modeling using Convolutional UNet architecture [3]. We first explore UNet with Attention gates [4] and then finally explore Diffusion Transformers [5]. All of these models are adapted and tuned to work for 1D ECG signals. We also employ a way to train the 2D models for signal generation. We would like to acknowledge the comprehensive tutorial on denoising processes by DeepFindr [6], which was endorsed by Andrej Karpathy.

2 Related Work

[7] uses multiple models of GANs, DCGANs WGANs and GANs with BiLSTM layers to name a few, to generate synthetic data. They measured the performance of the GANs models by comparing real and synthetic data using 1D metrics such as Dynamic Time Warping, Frechet Distance and Euclidean distances. In [8], the previous work builds upon it, implementing and comparing the performance of an improved Denoising Diffusion Probabilistic Models and Wasserstein GAN with Gradient Penalty. The authors converted the 1D signals to multi channel 2D representations before training with DDPM. However, WGAN-GP outperformed DDPM, which we hypothesize shows the complex temporal behaviour of ECG signals.

The biggest issue with these piece of work is that they only used data from the largest class (Normal class has around 90000 samples compared to the other classes which all have less than 10000 samples). Since the motive behind the project is to help removing bias in data samples, using the largest sample size class defeats the purpose entirely. We use the smaller sample classes (S, Q, F) along with N and see how our models perform.

[5] explores a new class of diffusion models by replacing the U-Net architecture with transformers that operate on latent image patches. The proposed architecture is scalable and outperforms prior diffusion models on ImageNet. We will be exploring the architecture and try to adapt it to 1D data for signal generation.

[9] proposes an autoregressive version of DDPM based on WaveNet [10]. The authors leverage Recurrent Neural Network mechanism to encode temporal data which is then used with DDPM.

[11] proposes a diffusion-based model specifically designed for vibration signal generation. It employs an improved U-net architecture with added TimeEmbedding blocks to enhance feature extraction from 1D time-series data such as single-frequency, multi-frequency, and bearing fault datasets.

In [12] The authors use conditional VAE and WGAN models to generate ECG signals taken from MITBIH dataset. For the experiments with GANs ,the 1D signal was first converted to 2D spectrograms using Short Time Fourier Transform. The generated spectrograms are converted back with ISTFT. VAE variants with 1D convolutions show better results than GANs. We are inspired by the 1D to 2D conversion and employ a stacking algorithm to evaluate 2D Unet and DiT diffusion models.

3 Methodology

3.1 Data and Preprocessing

The primary dataset used is the MIT-BIH Arrhythmia Database [13] provided by PhysioNet [14]. It is one of the most common benchmarks for ECG analysis. It includes 48 30-min two-channel ECG record studied by the BIH Arrhythmia Laboratory between 1975 and 1979. The recordings were digitized at 360 samples per second per channel with 11-bit resolution over a 10 mV range. Two or more cardiologists independently annotated each record. The dataset is highly imbalanced. The individual beats are divided into 4 main classes N ,F, V, S and 1 fabricated class Q.

- Normal (N): 82.8% of the total samples.
- Premature Ventricular Contraction (V): 6.6% of the total samples.
- Fusion of Ventricular and Normal (F): 0.7% of the total samples.
- Supraventricular premature beat (S): 2.5% of the total samples.

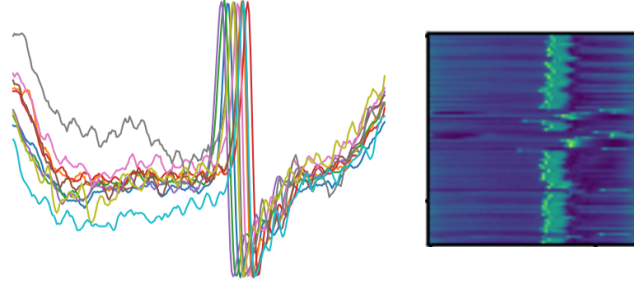


Figure 1: 1D and 2D ECG data

- Unclassifiable beat (Q): 7.3% of the total samples.

The beats in class "Q" do not follow a pattern as is the case in the other classes.

Each patient's data can be downloaded as a .mat file which can then be read in MATLAB. A script is implemented to read and view these individual samples after removing the baseline wander using high-pass Chebyshev filter. This approach would have been too time consuming as there was a need of individual PQRST beat complexes after collecting data from all the patients.

The segmented and consolidated data was kindly made available by [15]. Another script was implemented in MATLAB to process the structure and convert to python readable form as .csv files. Beat signals and corresponding label was then created in python and saved as a data frame in a pickle format. A separate data frame was created for each class of signals. To create the 2D images from the signals, we stacked the data and created (128,128) sized images. After model inference, we can extract each row to get back our 1D signals. Figure 1 shows a few examples of 1D and corresponding 2D data.

3.2 Diffusion Denoising Probabilistic Models

Denosing Diffusion Probabilistic Models (DDPM) [16] are a class of generative models that learn to generate data by reversing a fixed Markovian forward noising process. The forward process incrementally adds Gaussian noise to the input data, converting a clean sample $\mathbf{x}_0$ into a sequence $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\}$, where $\mathbf{x}_T$ approaches an isotropic Gaussian distribution.

The forward process is defined as:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}), \quad (1)$$

where $\beta_t \in (0, 1)$ is a predefined variance schedule. By composing these transitions, we get:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad (2)$$

with $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$.

The goal of DDPM is to learn a reverse process $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)), \quad (3)$$

where the mean and variance are parameterized by a neural network.

Figure 2 shows the forward and reverse process.

To simplify training, Ho et al. propose predicting the noise ϵ directly using a neural network ϵ_θ , optimizing the following objective:

$$L_{\text{simple}} = \mathbb{E}_{\mathbf{x}_0, t, \epsilon} \left[\left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, t) \right\|^2 \right]. \quad (4)$$

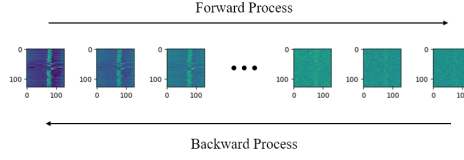


Figure 2: Diffusion process

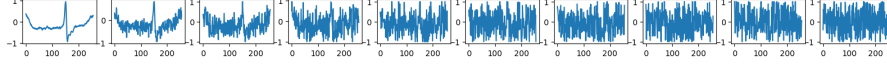


Figure 3: 1D denoising process on the ECG signals from one of our models.

This process is model agnostic. Thus we are free to use any kind of model that satisfies the input and output constraints. Let us take a look at those models. We used positional encoding and time embeddings in all models to give the inherent information of the time-series data and denoising step data respectively.

3.3 Baseline Model: UNet

The U-Net architecture [3] is widely used as the denoising backbone in diffusion models due to its ability to capture both local and global features effectively. Originally designed for biomedical image segmentation, U-Net consists of a symmetric encoder-decoder structure with skip connections. The encoder compresses the input through successive convolution and downsampling layers, extracting high-level features, while the decoder reconstructs the signal by progressively upsampling. Skip connections bridge corresponding layers in the encoder and decoder, preserving fine-grained details lost during downsampling. In the context of DDPM, U-Net is employed to model the reverse process $p_{\theta}(x_{t-1}|x_t)$ by predicting the added noise ϵ at each timestep.

For our implementation, the Unet architecture was configured with (64, 128, 256, 512, 1024) channel dimensions.

3.4 UNet with Attention Gates

To further enhance the representational power of U-Net, self-attention mechanisms are integrated into intermediate layers, enabling the model to capture long-range dependencies that convolutional layers alone struggle to represent. Attention blocks are incorporated into selected resolutions of the U-Net, allowing the model to dynamically weigh spatial or temporal regions of importance when generating or denoising signals. This is particularly beneficial in scenarios involving complex dependencies across time or space, such as in ECG signal generation where morphological consistency over long temporal spans is crucial. For our implementation, the Unet + Attention architecture was configured with (64, 128, 256, 512, 1024) channel dimensions, similar to UNet.

Figure 4 shows the UNet and UNet + Attention architecture

3.5 Diffusion Transformers

Diffusion Transformers replace the conventional convolutional U-Net backbone with a transformer architecture operating on latent patches. This approach leverages the scalability and modeling capabilities of transformers, particularly in capturing long-range dependencies. The diffusion processes are conducted in a compressed latent space rather than the high-dimensional data space. This integration significantly reduces computational overhead while maintaining high-quality generation. The DiT model was configured with patch size of 4, number of attention heads as 16 and number of blocks as 8

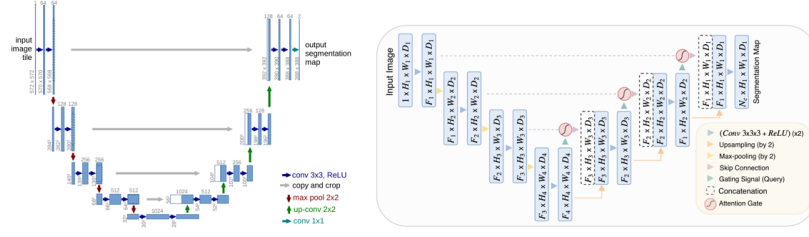


Figure 4: Left: Unet, Right: Unet with Attention gates

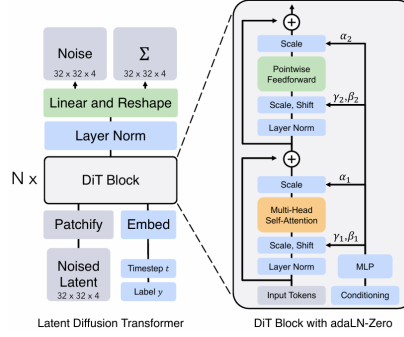


Figure 5: Diffusion Transformer Architecture

4 Experiments

4.1 Model Designs and Experiment Setting

DDPM allows for prediction of the noise ϵ added to an image at each step or the direct prediction of the clean image x_0 at a given timestep. The authors mention that predicting ϵ is easier and stable. Therefore, our DDPM training involves the prediction of ϵ . We used a linear noise scheduler for all the experiments. Table 4.1 shows the hyperparameters shared across all models to have consistent comparison.

Table 1: Hyperparameters

Noise Scheduler	T	β_0	β_T	Optimizer	Learning rate	Loss function	Epochs
Linear	300	0.0001	0.02	AdamW	1e-4	L2	501

4.2 Hypothesis

Our goal is to achieve models which can generate qualitatively and quantitatively similar synthetic samples compared to real signals. We want to use few samples from the classes like Q, S, F, etc. and see how well the models learn on less samples. This is key for supporting our motive of data augmentation.

4.3 Evaluation

4.3.1 Similarity metrics

Below table briefly describes the different metrics used based on [2] and [7], what their values mean and which model uses them.

Table 2: Metrics and their interpretation of values

Metric	Description	Interpretation of Value
UOI	The Universal Quality Index created by modeling an image distortion as a combination of loss of correlation, distortion of luminance, and contrast [17]	Higher is better
VIFP	Visual information fidelity is a full reference image quality assessment index based on the notion of image information extracted by the human visual system [18]	Higher is better
SAM	Spectral Angle Mapper (SAM) is a physically-based spectral classification that uses an n-D angle to match pixels to reference spectra [19]	Lower is better
ERGAS	Error relative global dimensionless synthesis is used to calculate the accuracy of Pan sharpened image considering normalized average error [20]	Lower is better
RASE	Relative Average Spectral Error is a metric used to evaluate the average performance of an image fusion method across different spectral bands [20]	Lower is better
SIFT	Scale-Invariant Feature Transform is a computer vision algorithm to detect, describe, and match local features in images [21]	Higher is better
SSIM	Structural similarity index is used for measuring image quality [22]	Higher is better
DTW	Dynamic Time Warping is an algorithm for measuring similarity between two temporal sequences, which may vary in speed [23],[24]	Lower is better

4.3.2 Comparisons

Using the similarity metrics provided above, we will be doing 4 comparisons for each class of signals. The ground truth samples selected, X , from the real dataset D are kept constant between all 4 comparisons: $X \sim D$

- Real signals randomly sampled $x \sim D$ vs. X
- Signals from $\hat{x}_{unet} \sim \theta_{unet}$ vs. X
- Signals from $\hat{x}_{unetAttn} \sim \theta_{unetAttn}$ vs. X
- Signals from $\hat{x}_{dit} \sim \theta_{dit}$ vs. X

5 Results

The model parameters and training time for each of the models were:

	UNet	UNet + Attention	DiT
Model Parameters	18,664,321	20,059,845	6,606,596
Training Time	02:05	02:11	09:31

5.1 Qualitative Evaluation

Let us look at a visual comparison between synthetic signals (blue) and real signals (orange) for the class 'S'.

5.1.1 UNet

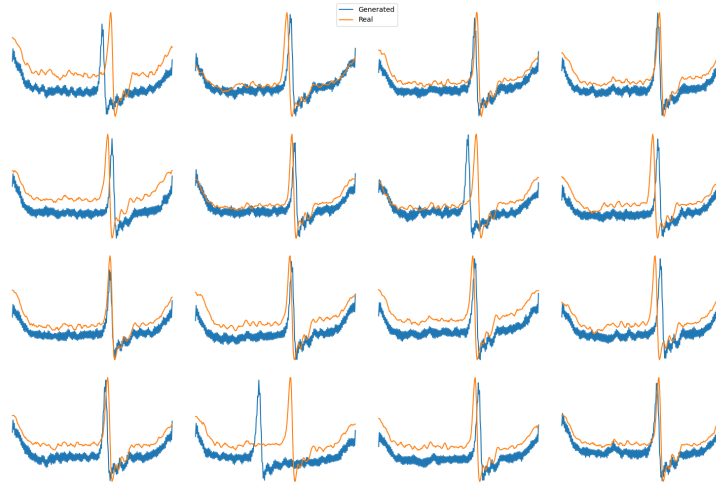


Figure 6: We can see the overall signal shape is matching but there is a lot of noise in the low voltage parts of the signal. This is intuitive as just convolutional layers would not be enough to capture small variations and undulations in the signal.

5.1.2 UNet + Attention

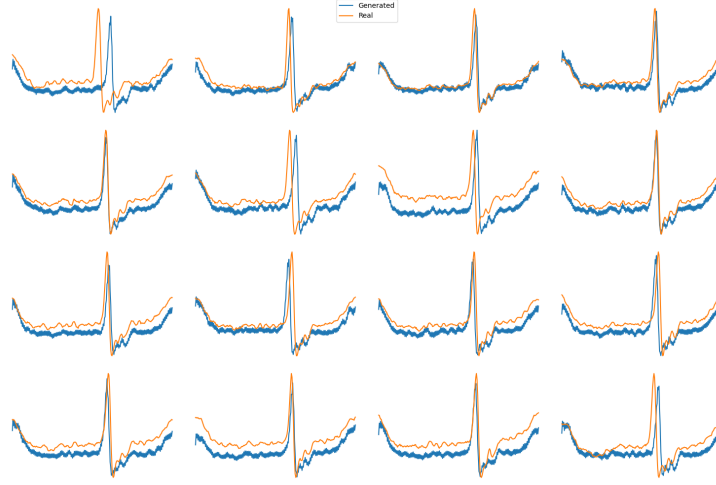


Figure 7: Unet with attention gates show less noise in the low voltage parts of the signal. The addition of self attention from the higher dimensional spaces allows better learning of the temporal data.

5.1.3 DiT

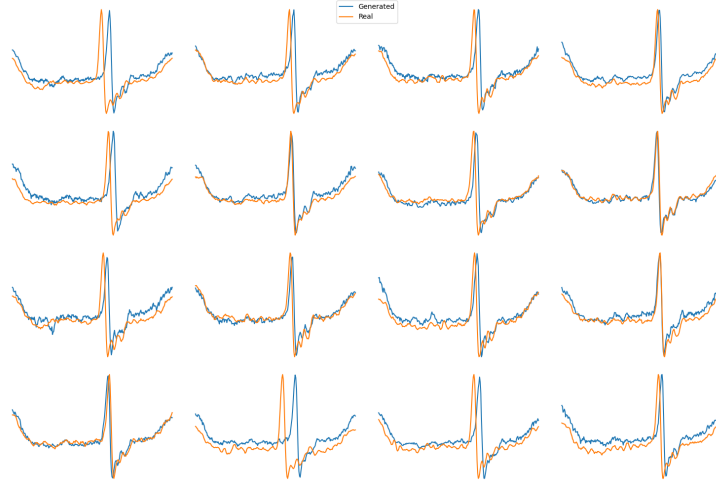


Figure 8: DiT shows the best performance visually. The synthetic signals show really good details and temporal structure.

5.2 Quantitative Evaluation: using metrics

Table 5.2 shows the different metric values for the synthetic and real signals from class 'S', as discussed in section 4.3.2. We can see that while UNet with Attention perform slightly better than our baseline model, DiT shows a massive improvement over both models. The values denote that synthetic signals from DiT are the closest to the real samples.

Table 3: Metrics for class S

Class 'S'	UQI	VIFP	SAM	ERGAS	RASE	SIFT	SSIM	DTW
UNet (Baseline)	0.976983	0.020747	0.140441	3.509854	310.643261	11.46875	0.910642	33.832653
UNet + Attention	0.98397	0.014356	0.123602	3.090588	261.918316	16.40625	0.915315	32.251479
DiT	0.991198	0.027133	0.098887	2.476627	198.162326	21.734375	0.930084	10.942024
Real	0.992074	0.041272	0.097069	2.432174	195.699083	28.171875	0.929843	9.948521

Table 5.2 shows the different metric values for the synthetic and real signals from class 'Q'. Since 'Q' is a fabricated class with unknown samples put in this group, the performance is a little worse overall for all three models. DiT still outperforms the baseline, but Unet with Attention is not able to perform as well. This shows that DiT models can learn various different representations and produce varied samples.

Table 4: Metrics for class Q

Class 'Q'	UQI	VIFP	SAM	ERGAS	RASE	SIFT	SSIM	DTW
UNet (Baseline)	0.981467	0.021026	0.127573	3.189263	269.653488	11.484375	0.919327	22.883154
UNet + Attention	0.972355	0.020376	0.146466	3.663008	321.165366	4.53125	0.91484	28.608248
DiT	0.991194	0.014725	0.099059	2.48173	201.849748	19.046875	0.928326	45.942745
Real	0.991618	0.032211	0.097265	2.436973	195.198826	21.15625	0.931696	17.275514

6 Experimental: 2D training

We implemented the standard 2D UNet and Diffusion Transformer models as well to see how well it learns signal data. For this process to work we stacked multiple 1D signals to form an image. These images were then used in the diffusion process. To extract the signals, we just extract each row of the image and store it.

We could not see good results with either of Unet 2D and DiT 2D models. It is likely that the models, especially the transformer model, would require more epochs to learn. Such images would also be difficult for Unets to learn as there are a lot of tiny details in such images, as can be seen on the right image of Figure 1. Figure 9 shows the results of training with 2D data for only 501 epochs for the UNet (left) and DiT (right). [ht]

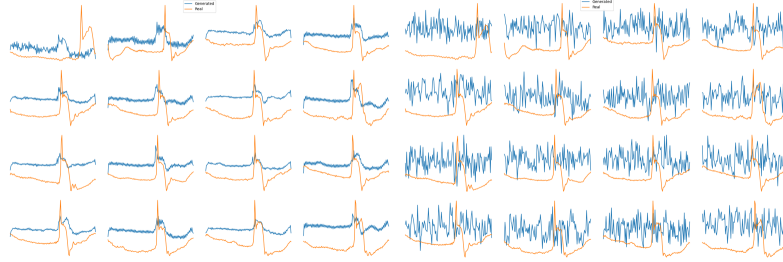


Figure 9: Output of 1D data training for class Q

7 Conclusion

We have seen how Attention layers can enhance the output of the denoising process for 1D temporal biosignals, especially within the architecture of Transformers. The DiT model adapted for 1D signals is scalable and can produce excellent results which can help us achieve the goals of data augmentation and creating balanced datasets for use in the healthcare field, without compromising security and privacy. Future work includes implementing the class embeddings to help create just one model for any kind of ECG signal class instead of having to create one model for each class. This is not a big issue as we saw how powerful these models are despite being lightweight in size and having respectable training speeds. We could use these synthetic signals and curate a dataset to then run well known classification models to further evaluate the generation models.

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